

SQL-Based Analysis of Skin Cancer Lesions



10ALYTICS DATA SCIENCE PROGRAM

Pod Polaris



PROJECT OVERVIEW

Background, Business Problem & Objectives



BACKGROUND

Skin cancer is one of the most common cancers globally. Early detection through data-driven diagnostics can significantly improve patient outcomes.

The DermAI Diagnostics dataset provides clinical patient records and lesion data for analysis using structured SQL queries.



BUSINESS PROBLEM

Healthcare providers lack structured insights into skin lesion prevalence, demographic risk factors, and diagnostic patterns.

Without data analysis, clinical decisions rely on anecdotal evidence rather than evidence-based patterns derived from patient data.



OBJECTIVES

- Develop a relational SQL database for skin cancer analysis.
- Examine demographic and environmental factors associated with skin lesion diagnoses.
- Analyze lesion characteristics to identify clinically relevant patterns within the dataset.
- Create a machine-learning-ready dataset by integrating patient and lesion information into a unified structure.
- Explore the potential application of the dataset for artificial intelligence, predictive analytics, and dermatological research.

DATASET OVERVIEW

DermAI Diagnostics Database Schema

patient_info

Column	Description
patient_id	Unique patient identifier (PK)
age	Patient age at diagnosis
gender	Male / Female
smoking	Smoking status (boolean)
pesticide	Pesticide exposure (boolean)

lesion_info

Column	Description
lesion_id	Unique lesion identifier (PK)
patient_id	Foreign key to patient_info
diagnostic	Lesion type / diagnosis
region	Body region of lesion
dx_length / dx_width	Lesion dimensions
biopsied	Biopsy confirmation status

patient_id is the linking key between tables

DATA PREPARATION

Renaming Tables for Clarity

```
ALTER TABLE table1 RENAME TO patient_info;  
ALTER TABLE table2 RENAME TO lesion_info;
```

BEFORE

table1
table2

WHY RENAME?

- Renamed generic table names to meaningful business names
- Improves code readability and maintainability
- Aligns database objects with domain terminology
- Makes SQL queries self-documenting for team collaboration

AFTER

patient_info
lesion_info

RATIONALE

Renaming generic table names to meaningful business names improves readability, simplifies query development, and enhances database maintainability.

VERIFYING TABLES

Confirming Database Objects

```
SELECT table_name
FROM information_schema.tables
WHERE table_schema = 'public';
```

QUERY PURPOSE

This query was used to verify that the renamed tables (patient_info and lesion_info) were successfully created and exist within the public schema of the PostgreSQL database before analysis began.

WHY THIS STEP MATTERS

Verifying database objects before running analysis ensures that all required tables exist and are correctly named. This reduces the risk of query failures and improves the reliability of the analytical results.

RESULT

table_name
patient_info
lesion_info



Tables verified

EXPLORING PATIENT_INFO

Understanding Available Patient Fields

```
SELECT * FROM patient_info LIMIT 3;
```

CLAUSE BREAKDOWN

SELECT *

Returns all columns from the table

FROM patient_info

Specifies the source table

LIMIT 3

Restricts output to first 3 rows for quick inspection

SAMPLE OUTPUT

patient_id	age	gender	smoking	pesticide
PAT001	45	Male	TRUE	FALSE
PAT002	62	Female	FALSE	TRUE
PAT003	38	Male	FALSE	FALSE

PURPOSE

Displays sample patient records to understand data structure

Identifies available fields for downstream analysis

Validates data types and completeness before running complex queries

EXPLORING LESION_INFO

Understanding Available Lesion Fields

```
SELECT * FROM lesion_info LIMIT 3;
```

CLAUSE BREAKDOWN

SELECT *

Returns all columns from the lesion table

FROM lesion_info

Specifies the lesion information table

LIMIT 3

Quick preview of first 3 lesion records

SAMPLE OUTPUT

lesion_id	patient_id	diagnostic	region	biopsied
LES001	PAT001	ACK	FACE	TRUE
LES002	PAT002	MEL	BACK	TRUE
LES003	PAT003	BCC	SCALP	FALSE

PURPOSE

Reveals lesion-level detail including diagnosis type and body region

Confirms lesion dimensions and biopsy status fields are available

Validates foreign key relationship through patient_id

SQL ANALYSIS

METHODOLOGY

8 structured queries linked to project objectives

AIM 1: LESION PREVALENCE ANALYSIS

Query 1

- Most Common Skin Lesion

AIM 2: DEMOGRAPHIC & ENVIRONMENTAL RISK FACTORS

Query 2

- Lesions by Gender

Query 3

- Average Age by Diagnosis

Query 4

- Smoking & Lesion Types

Query 5

- Pesticide Exposure & Diagnosis

AIM 3: LESION CHARACTERISTICS & CLINICAL PATTERNS

Query 6

- Most Affected Body Region

Query 7

- Average Lesion Size by Diagnosis

Query 8

- Biopsy Confirmation

QUERY 1: MOST COMMON SKIN LESION

Which lesion type occurs most frequently?

```
SELECT diagnostic, COUNT(*) AS total_cases
FROM lesion_info
GROUP BY diagnostic
ORDER BY total_cases DESC;
```

SYSTEMATIC EXPLANATION

SELECT diagnostic

Retrieves the lesion type from each record

COUNT(*) AS total_cases

Counts rows per group; alias improves readability

GROUP BY diagnostic

Groups all records by lesion type

ORDER BY total_cases DESC

Sorts results with highest frequency first

CATEGORY:

Lesion Prevalence Analysis (Aim 1)

WHY THIS QUERY?

This query was used to identify the most prevalent skin lesion diagnosis within the dataset. Understanding lesion prevalence helps prioritize screening efforts and highlights the lesion types that occur most frequently in the study population.

QUERY 1: RESULTS & INSIGHT

Most Common Skin Lesion

461

ACK Cases

#1

Most Prevalent



KEY FINDING

ACK (Actinic Keratosis) is the most prevalent lesion type with 461 cases, indicating a high burden of pre-cancerous skin conditions in the patient population.



RECOMMENDATION

Strengthen screening programs for ACK. Early detection of pre-cancerous lesions can prevent progression to squamous cell carcinoma.

Diagnostic	Total Cases
ACK	461
BCC	273
NEV	144
SEK	137
SCC	56
MEL	17

QUERY 2: LESIONS BY GENDER

Are certain lesions more common in males or females?

```
SELECT p.gender, l.diagnostic, COUNT(*) AS total_cases
FROM patient_info p
JOIN lesion_info l
  ON p.patient_id = l.patient_id
GROUP BY p.gender, l.diagnostic
ORDER BY p.gender, total_cases DESC;
```

SYSTEMATIC EXPLANATION

SELECT p.gender, l.diagnostic

Retrieves the patient's gender and lesion diagnosis.

COUNT(*)

Counts the total number of cases for each gender-diagnosis combination.

JOIN patient_info & lesion_info

Combines patient records with lesion information.

GROUP BY p.gender, l.diagnostic

Groups cases by gender and diagnosis type.

ORDER BY p.gender, total_cases DESC

Ranks diagnoses by frequency within each gender group.

CATEGORY:

Demographic Risk Factor Analysis (Aim 2)

WHY THIS QUERY?

This query was used to determine whether certain lesion diagnoses occur more frequently in males or females. Gender is a key demographic factor that may influence disease occurrence.

QUERY 2: RESULTS & INSIGHT

Lesions by Gender

Gender	Diagnostic	Total Cases
Female	ACK	135
Female	BCC	122
Female	NEV	42
Female	SEK	28
Female	SCC	25
Female	MEL	10
Male	ACK	326
Male	BCC	151
Male	SEK	109
Male	NEV	102
Male	SCC	31
Male	MEL	7



KEY FINDING

Analysis of gender distribution across diagnostic categories showed that males had a higher number of lesions than females in most diagnosis groups. ACK was the most common diagnosis among both males and females, with 326 male cases and 135 female cases. These findings suggest a higher burden of skin lesions among male patients within the dataset.



RECOMMENDATION

Further investigation should be conducted to determine the underlying factors contributing to the higher lesion prevalence observed among male patients.

QUERY 3: AVERAGE AGE BY DIAGNOSIS

What is the average patient age for each lesion type?

```
SELECT l.diagnostic,  
       ROUND(AVG(p.age)::numeric, 1) AS avg_age  
FROM patient_info p  
JOIN lesion_info l  
  ON p.patient_id = l.patient_id  
GROUP BY l.diagnostic  
ORDER BY avg_age DESC;
```

SYSTEMATIC EXPLANATION

SELECT l.diagnostic

Retrieves each lesion diagnosis.

AVG(p.age)

Calculates the average patient age for each diagnosis.

ROUND(..., 1)

Rounds the result to one decimal place.

JOIN patient_info & lesion_info

Combines patient and lesion data.

GROUP BY l.diagnostic

Groups records by diagnosis type.

ORDER BY avg_age DESC

Ranks diagnoses from oldest to youngest average age.

CATEGORY:

Demographic Risk Factor Analysis (Aim 2)

WHY THIS QUERY?

This query was used to examine the average age associated with each diagnosis. The results help identify age groups that may be more vulnerable to specific lesion types.

QUERY 3: RESULTS & INSIGHT

Average Age by Diagnosis

Diagnostic	Average Age
SCC	68.5
SEK	64.0
BCC	62.3
ACK	61.0
MEL	59.8
NEV	34.3

68.5

SCC Highest Avg Age

34.3

NEV Lowest Avg Age



KEY FINDING

SCC has the highest average age (68.5), suggesting older patients are more susceptible. NEV stands out with the lowest average age (34.3), indicating it affects a distinctly younger population.



Recommendation

Implement age-stratified screening protocols, with enhanced monitoring for patients over 60.

QUERY 4: SMOKING & LESION TYPES

Does smoking correlate with specific lesion types?

```
SELECT p.smoke,  
       l.diagnostic,  
       COUNT(*) AS total_cases  
FROM patient_info p  
JOIN lesion_info l  
ON p.patient_id = l.patient_id  
GROUP BY p.smoke, l.diagnostic  
ORDER BY total_cases DESC;
```

SYSTEMATIC EXPLANATION

SELECT p.smoke, l.diagnostic

Retrieves smoking status and lesion diagnosis.

COUNT(*) AS total_cases

Counts cases in each category.

JOIN patient_info & lesion_info

Links patient records with lesion data.

GROUP BY p.smoke, l.diagnostic

Groups cases by smoking status and diagnosis.

ORDER BY total_cases DESC

Displays the most frequent combinations first.

CATEGORY:

Environmental & Lifestyle Risk Factor Analysis (Aim 2)

WHY THIS QUERY?

This query was used to explore whether smoking status is associated with particular lesion diagnoses and assess smoking as a potential risk factor.

QUERY 4: RESULTS & INSIGHT

Smoking and Lesion Types

Smoking Status	Diagnostic	Total Cases
False	ACK	448
False	BCC	243
False	NEV	143
False	SEK	136
False	SCC	41
False	MEL	15
True	BCC	30
True	SCC	15
True	ACK	13
True	MEL	2
True	SEK	1
True	NEV	1



KEY FINDING

Analysis of smoking status revealed that most lesion cases were recorded among non-smokers. Among smokers, Basal Cell Carcinoma (BCC) and Squamous Cell Carcinoma (SCC) were the most common diagnoses. While the dataset suggests a possible association between smoking and certain lesion types, additional statistical analysis would be required to establish a definitive relationship.



RECOMMENDATION

Integrate smoking cessation guidance into dermatology screening programs as a complementary preventive measure.

QUERY 5: PESTICIDE EXPOSURE & DIAGNOSIS

Does pesticide exposure correlate with skin lesion types?

```
SELECT p.pesticide, l.diagnostic,  
       COUNT(*) AS total_cases  
FROM patient_info p  
JOIN lesion_info l  
  ON p.patient_id = l.patient_id  
GROUP BY p.pesticide, l.diagnostic  
ORDER BY p.pesticide, total_cases DESC;
```

SYSTEMATIC EXPLANATION

SELECT p.pesticide, l.diagnostic

Retrieves pesticide exposure status and lesion diagnosis.

COUNT(*)

Counts the total number of cases for each pesticide-diagnosis combination.

JOIN patient_info & lesion_info

Combines patient information with lesion records.

GROUP BY p.pesticide, l.diagnostic

Groups cases by pesticide exposure and diagnosis type.

ORDER BY p.pesticide, total_cases DESC

Ranks diagnoses by frequency within each pesticide exposure group.

CATEGORY:

Environmental Risk Factor Analysis (Aim 2)

WHY THIS QUERY?

This query was used to investigate whether exposure to pesticides is associated with differences in lesion occurrence and diagnosis patterns.

QUERY 5: RESULTS & INSIGHT

Pesticide Exposure and Diagnosis

Pesticide Exposure	Diagnostic	Total Cases
False	ACK	399
False	BCC	146
False	NEV	141
False	SEK	135
False	SCC	34
False	MEL	10
True	BCC	127
True	ACK	62
True	SCC	22
True	MEL	7
True	NEV	3
True	SEK	2



KEY FINDING

Among patients exposed to pesticides, **BCC is the most common diagnosis**, with **127 cases**. This is different from the non-exposed group, where **ACK is the most common**, with **399 cases**.



RECOMMENDATION

Individuals with occupational or environmental exposure to pesticides should be encouraged to use protective equipment and undergo regular skin examinations to facilitate early detection of potentially harmful lesions.

QUERY 6: MOST AFFECTED BODY REGION

Which body region has the highest lesion frequency?

```
SELECT region, COUNT(*) AS total_cases
FROM lesion_info
GROUP BY region
ORDER BY total_cases DESC;
```

SYSTEMATIC EXPLANATION

SELECT region

Retrieves the body region field from lesion records

COUNT(*) AS total_cases

Counts lesions per region

GROUP BY region

Creates frequency distribution by body area

ORDER BY total_cases DESC

Ranks regions from most to least affected

CATEGORY:

Lesion Characteristics Analysis (Aim 3)

WHY THIS QUERY?

This query was used to identify the anatomical regions where skin lesions occur most frequently. Understanding lesion distribution supports clinical assessment and early detection efforts.

QUERY 6: RESULTS & INSIGHT

Most Affected Body Region

Region	Total Cases
FACE	278
FOREARM	219
CHEST	124
BACK	105
ARM	92
NOSE	68
HAND	55
NECK	37
THIGH	35
EAR	32
ABDOMEN	20
SCALP	11
LIP	7
FOOT	5



KEY FINDING

The face was the most frequently affected body region with 278 cases, followed by the forearm (219 cases) and chest (124 cases). These regions accounted for the highest number of recorded lesions in the dataset.



RECOMMENDATION

Increase awareness of skin health and encourage regular examination of body regions with higher lesion occurrence, particularly the face, forearm, and chest, to support early detection and timely medical evaluation.

QUERY 7: AVERAGE LESION SIZE BY DIAGNOSIS

Which lesion types are largest on average?

```
SELECT diagnostic,  
       ROUND(AVG((diameter_1 + diameter_2)/2)::numeric, 2) AS avg_lesion_size  
FROM lesion_info  
GROUP BY diagnostic  
ORDER BY avg_lesion_size DESC;
```

SYSTEMATIC EXPLANATION

SELECT diagnostic

Retrieves each lesion diagnosis category.

(diameter_1 + diameter_2) / 2

Calculates the average size of each lesion using its two diameters.

AVG(...)

Computes the average lesion size for each diagnosis.

ROUND(..., 2)

Rounds the result to two decimal places.

GROUP BY diagnostic

Groups lesions by diagnosis type.

ORDER BY avg_lesion_size DESC

Ranks diagnoses from largest to smallest average lesion size.

CATEGORY:

Lesion Characteristics Analysis (Aim 3)

WHY THIS QUERY?

This query was used to compare lesion sizes across diagnoses and identify lesion types that tend to present with larger dimensions.

QUERY 7: RESULTS & INSIGHT

Average Lesion Size by Diagnosis

Diagnostic	Average Lesion Size
MEL	14.09
SCC	10.46
BCC	9.97
ACK	2.62
SEK	0.80
NEV	0.67



KEY FINDING

MEL (Melanoma) has the largest average size (14.09mm), which is clinically significant as larger melanomas often indicate later-stage diagnosis and poorer prognosis.



RECOMMENDATION

Focus early detection efforts on melanoma. Smaller size at diagnosis correlates with better survival outcomes.

QUERY 8: BIOPSY CONFIRMATION

How many lesions were confirmed via biopsy?

```
SELECT biopsed,  
       COUNT(*) AS total_cases  
FROM lesion_info  
GROUP BY biopsed;
```

SYSTEMATIC EXPLANATION

SELECT biopsed

Retrieves the biopsy status of each lesion.

COUNT(*) AS total_cases

Counts the number of lesions in each biopsy group.

GROUP BY biopsed

Groups lesions based on whether a biopsy was performed.

CATEGORY:

Clinical Validation Analysis (Aim 3)

WHY THIS QUERY?

This query was used to evaluate the extent of biopsy confirmation within the dataset and assess the level of clinical verification of lesion diagnoses.

QUERY 8: RESULTS & INSIGHT

Biopsy Confirmation

458

Biopsies Confirmed

Biopsied	Total Cases
False	630
True	458



KEY FINDING

Analysis of biopsy status revealed that 458 lesions (42.1%) were biopsy-confirmed, while 630 lesions (57.9%) were diagnosed without biopsy confirmation. This indicates that a substantial proportion of lesions were evaluated using other methods



RECOMMENDATION

Investigate why ~58% of lesions lack biopsy confirmation. Implement follow-up protocols to close this significant diagnostic gap.

SUMMARY OF KEY FINDINGS

Summary of All Analysis Results

Analysis Area	Result
Most Common Lesion	ACK (461 cases)
Gender Pattern	Males more affected
Highest Average Age	SCC (68.5 years)
Smoking Impact	Proportional elevation in smokers
Pesticide Impact	Elevated rates in exposed patients
Most Affected Region	FACE (278 cases)
Largest Lesion Type	MEL (Melanoma)
Biopsy Confirmed	458 (42%)

RECOMMENDATIONS

Data-Driven Actions for Healthcare Improvement

01 ACK Screening

Prioritize screening programs for Actinic Keratosis, the most prevalent pre-cancerous condition

02 Gender-Targeted Outreach

Implement male-focused awareness campaigns for skin cancer prevention and early detection

03 Age-Stratified Protocols

Enhanced monitoring for patients over 60, particularly for SCC and ACK

04 Lifestyle Integration

Include smoking cessation and pesticide exposure assessments in dermatology workflows

05 Early Detection

Encourage regular examination of body regions with higher lesion occurrence, particularly the face and forearm to support early detection and timely medical evaluation.

06 Biopsy Gap Closure

Investigate and resolve the ~58% unconfirmed biopsy gap to strengthen diagnostics

MACHINE LEARNING-READY DATASET CREATION

```
CREATE TABLE ml_dataset AS
SELECT p.patient_id, p.age, p.gender,
       p.smoke, p.pesticide,
       l.diagnostic, l.region,
       ROUND(((l.diameter_1+l.diameter_2)/2)::numeric,2) AS lesion_size,
       l.biopsed
FROM patient_info p
JOIN lesion_info l ON p.patient_id = l.patient_id;
```

WHAT WAS DONE?

Combined Data

Combined patient demographics and lesion information into a single structured dataset.

Joined Tables

Joined patient_info and lesion_info using patient_id.

Key Variables

Selected key variables relevant for machine learning.

New Table

Created a new table called ml_dataset.

ML-Ready Format

Prepared data in a format suitable for future predictive analytics and AI model development.

RESULT

A consolidated dataset containing demographic, environmental, and lesion-specific features for each patient.

SAMPLE OUTPUT OF ml_dataset

Structured Data Ready for AI Applications

```
Select *  
from ml_dataset  
limit 10;
```

patient_id	age	gender	smoke	pesticide	diagnostic	region	lesion_size	biopsed
PAT_100	14	MALE	FALSE	FALSE	NEV	FACE	5.00	TRUE
PAT_1000	65	MALE	FALSE	FALSE	SEK	EAR	0.00	FALSE
PAT_1006	63	MALE	FALSE	FALSE	ACK	FOREARM	0.00	FALSE
PAT_1008	64	MALE	FALSE	FALSE	SEK	FACE	0.00	FALSE
PAT_101	59	MALE	TRUE	FALSE	BCC	NECK	17.50	TRUE
PAT_1013	43	MALE	FALSE	FALSE	ACK	BACK	0.00	FALSE
PAT_1014	41	MALE	FALSE	FALSE	ACK	FOREARM	0.00	FALSE
PAT_1017	67	MALE	FALSE	FALSE	ACK	FACE	0.00	FALSE
PAT_1018	37	MALE	FALSE	FALSE	ACK	ARM	0.00	FALSE
PAT_1019	48	MALE	FALSE	FALSE	ACK	ARM	0.00	FALSE

DATASET SUMMARY

- **9 Features** — Per Patient
- **Demographic** — Age, Gender
- **Environmental** — Smoking, Pesticide Exposure
- **Clinical** — Region, Lesion Size
- **Diagnostic Outcome** — Diagnosis, Biopsy Status
- **Ready for** — Supervised Machine Learning

Note: This query was used to verify that the machine-learning-ready dataset was successfully created.

POTENTIAL AI APPLICATIONS

From Structured Data to AI-Driven Healthcare Solutions

EXAMPLE 1

DIAGNOSIS PREDICTION

INPUT

Age: **68**
Gender: **Male**
Region: **FACE**
Lesion Size: **10.5 mm**
Smoking: **Yes**

AI PREDICTION: SCC

Purpose: Predict the most likely lesion diagnosis using patient and lesion characteristics.

EXAMPLE 2

BIOPSY RECOMMENDATION

INPUT

Age: **72**
Region: **FACE**
Lesion Size: **14 mm**
Diagnosis: **MEL**
Smoking: **—**

AI PREDICTION: BIOPSY: YES

Purpose: Assist clinicians in identifying lesions that may require further investigation.

POTENTIAL AI APPLICATIONS

From Structured Data to AI-Driven Healthcare Solutions

EXAMPLE 3

HIGH-RISK PATIENT ID

INPUT

Age: **70**

Gender: **Male**

Risk Profile: **Yes**

Pesticide Exposure: **Yes**

Smoking: **Elevated**

AI PREDICTION: HIGH-RISK PATIENT

Purpose: Support preventive screening and early intervention for at-risk patients.

EXAMPLE 4

LESION LOCATION PREDICTION

INPUT

Age: **65**

Gender: **Male**

Lesion Size: **ACK**

Diagnosis: **8.0 mm**

Smoking: **—**

AI PREDICTION: REGION: FACE

Purpose: Support research into lesion distribution patterns and disease presentation.

CONCLUSION

Project Summary & Impact

This project successfully used SQL to analyze skin lesion data and identify patterns in diagnoses, demographics, risk factors, lesion characteristics, and biopsy status. The findings showed that ACK was the most common lesion type, melanoma had the largest average lesion size, and lesions were more frequently observed among males, older adults, and in the face and forearm regions. Patterns related to smoking status, pesticide exposure, and biopsy confirmation provided additional insights into factors associated with skin lesions. Overall, the project demonstrates the value of SQL in supporting healthcare data analysis, dermatological research, and future machine learning applications.

THANK YOU



Pod Polaris

10Alytics Data Scientist Program

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